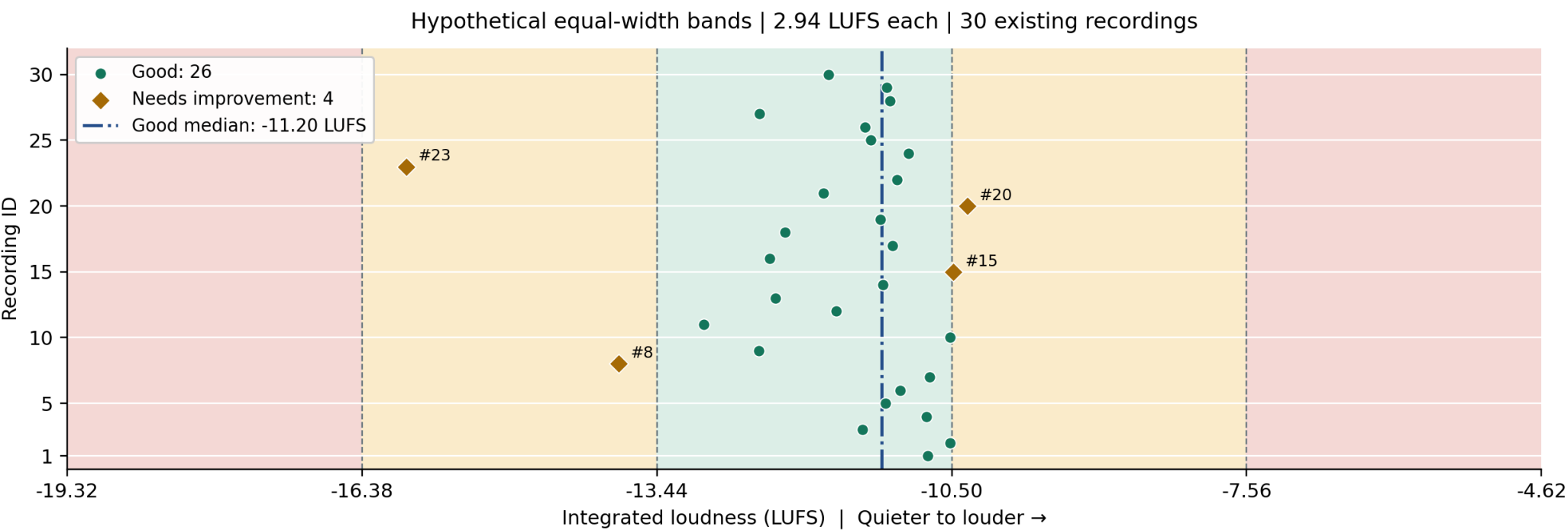


Loudness: equal-width theoretical ranges

KaraOK / Hypothetical sampling scenario / 30 existing recordings / 18 September 2026

Leftmost - Bad	Left - Needs improvement	Good	Right - Needs improvement	Rightmost - Bad
-19.32 to < -16.38 LUFS	-16.38 to < -13.44 LUFS	-13.44 to -10.50 LUFS	> -10.50 to -7.56 LUFS	> -7.56 to -4.62 LUFS

Five displayed bands, each 2.94 LUFS wide. Values below -19.32 or above -4.62 LUFS also remain Bad in this scenario.



Under these hypothetical boundaries: 26 Good / 4 Needs improvement / 0 Bad

Needs improvement: recordings 8 and 23 on the low side; 15 and 20 on the high side. IDs are not time order.

Good-data median: -11.20090 LUFS, calculated from the 26 recordings inside the Good range; not the range midpoint.

Good includes both endpoints. -16.38 and -7.56 belong to Needs improvement; values beyond those outer cutoffs are Bad.

All 30 source recordings were labeled good. Equal spacing is a sampling hypothesis, not a validated quality rule.

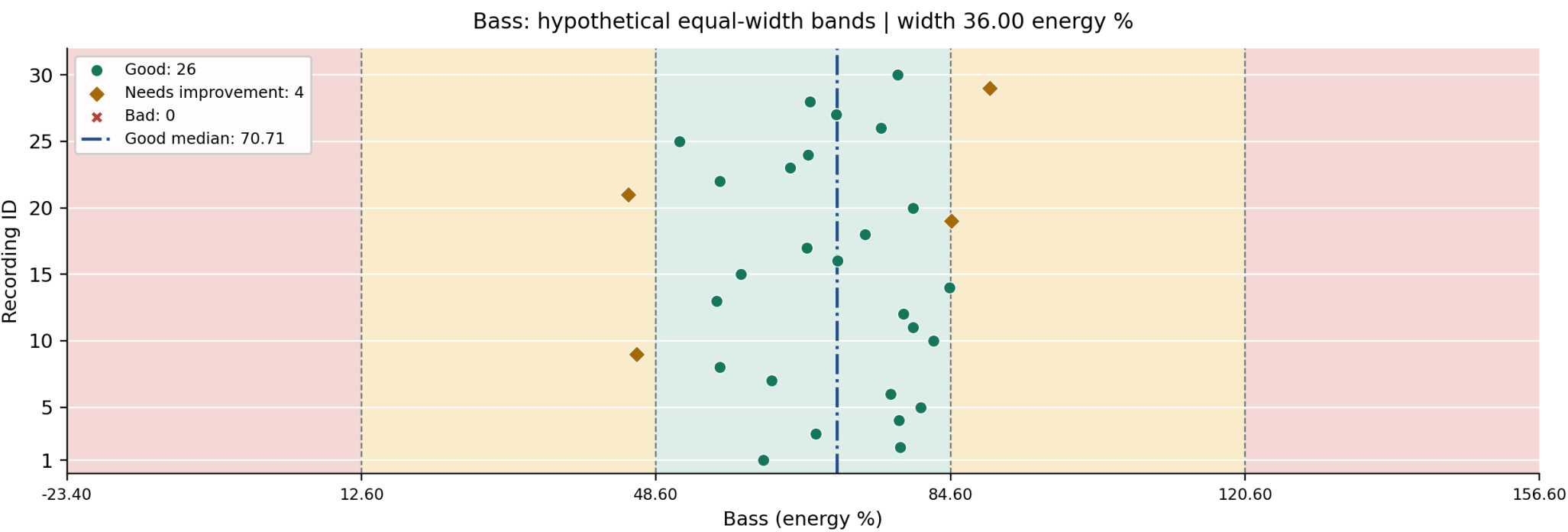
Production thresholds are unchanged. Collect new recordings across these bands and use independent listener ratings to validate them.

Bass: equal-width theoretical ranges

KaraOK / Hypothetical sampling scenario / 30 existing recordings / Units: energy %

Leftmost - Bad	Left - Needs improvement	Good	Right - Needs improvement	Rightmost - Bad
-23.40 to < 12.60	12.60 to < 48.60	48.60 to 84.60	> 84.60 to 120.60	> 120.60 to 156.60

Five equal-width theoretical bands. The graph and counts use full-precision thresholds; displayed values are rounded.



Under these hypothetical boundaries: 26 Good / 4 Needs improvement / 0 Bad

Good-data median: 70.71 energy %, from the 26 recordings inside the Good range; not the range midpoint.

Valid measurement domain: 0 to 100 energy %. Equal-width extensions outside the valid domain are not collection targets.

Good includes both endpoints; the adjacent outer cutoffs belong to Needs improvement. Further tails are Bad in this scenario.

0 recordings lie outside the displayed five-band window; arrows show their IDs and values. All 30 enter the counts.

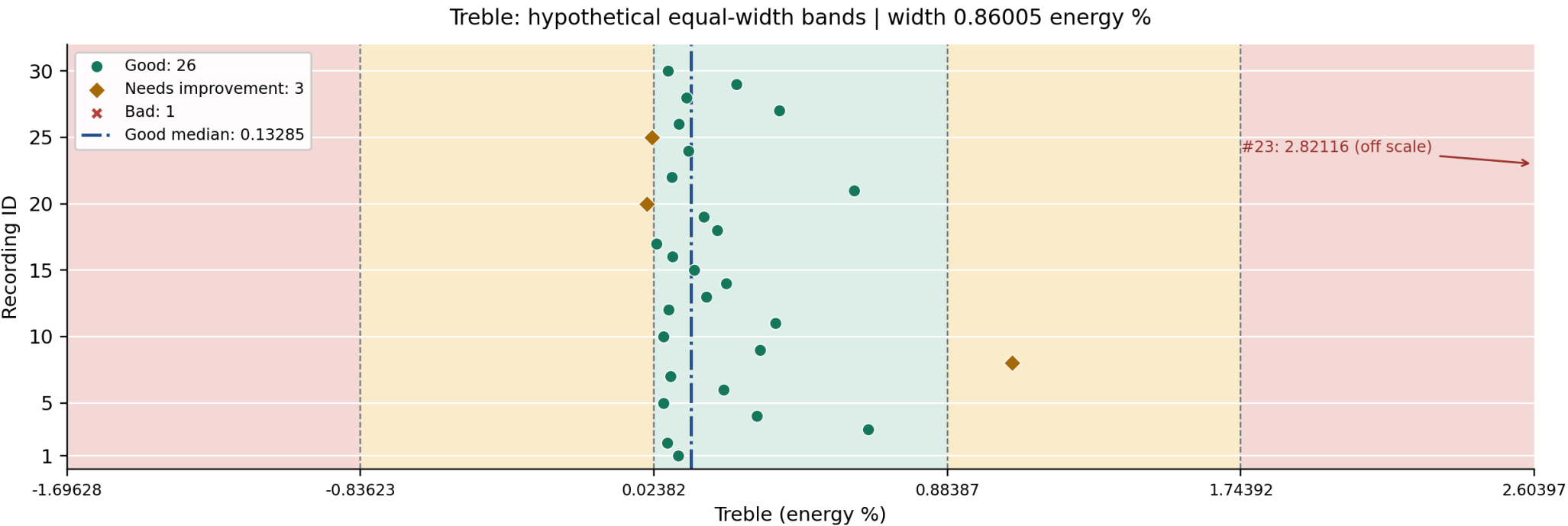
All source recordings were labeled good. These hypothetical classes need independent listener validation; production thresholds are unchanged.

Treble: equal-width theoretical ranges

KaraOK / Hypothetical sampling scenario / 30 existing recordings / Units: energy %

Leftmost - Bad	Left - Needs improvement	Good	Right - Needs improvement	Rightmost - Bad
-1.69628 to < -0.83623	-0.83623 to < 0.02382	0.02382 to 0.88387	> 0.88387 to 1.74392	> 1.74392 to 2.60397

Five equal-width theoretical bands. The graph and counts use full-precision thresholds; displayed values are rounded.



Under these hypothetical boundaries: 26 Good / 3 Needs improvement / 1 Bad

Good-data median: 0.13285 energy %, from the 26 recordings inside the Good range; not the range midpoint.

Valid measurement domain: 0 to 100 energy %. Equal-width extensions outside the valid domain are not collection targets.

Good includes both endpoints; the adjacent outer cutoffs belong to Needs improvement. Further tails are Bad in this scenario.

1 recordings lie outside the displayed five-band window; arrows show their IDs and values. All 30 enter the counts.

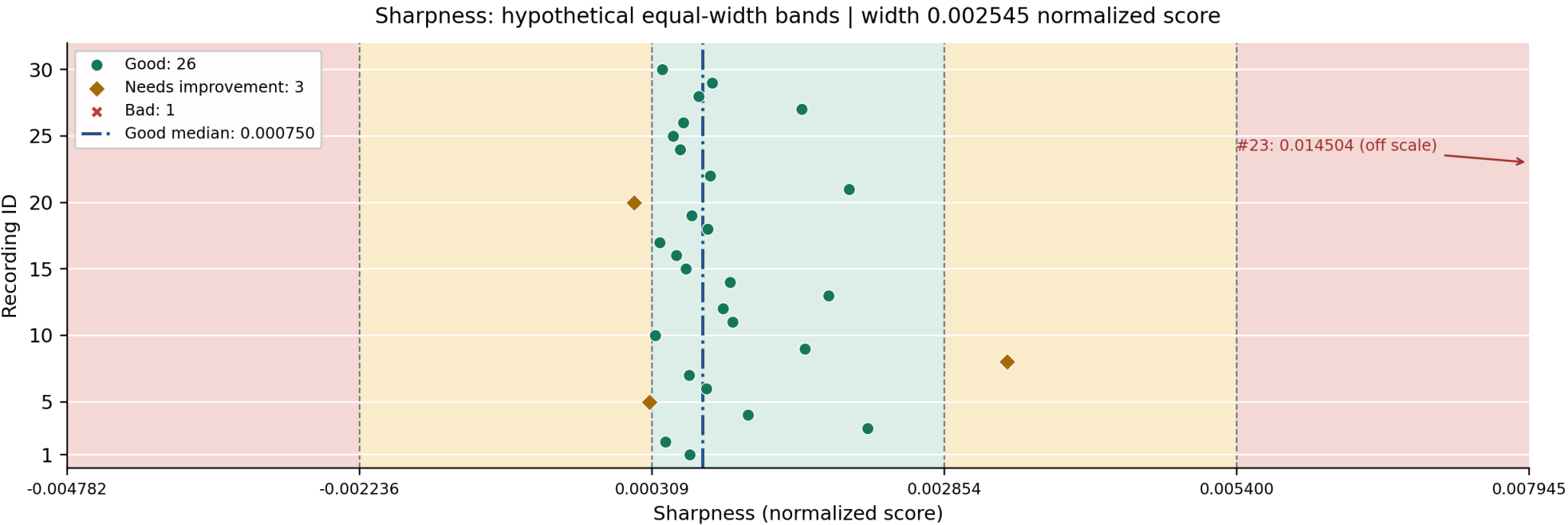
All source recordings were labeled good. These hypothetical classes need independent listener validation; production thresholds are unchanged.

Sharpness: equal-width theoretical ranges

KaraOK / Hypothetical sampling scenario / 30 existing recordings / Units: normalized score

Leftmost - Bad	Left - Needs improvement	Good	Right - Needs improvement	Rightmost - Bad
-0.004782 to < -0.002236	-0.002236 to < 0.000309	0.000309 to 0.002854	> 0.002854 to 0.005400	> 0.005400 to 0.007945

Five equal-width theoretical bands. The graph and counts use full-precision thresholds; displayed values are rounded.



Under these hypothetical boundaries: 26 Good / 3 Needs improvement / 1 Bad

Good-data median: 0.000750 normalized score, from the 26 recordings inside the Good range; not the range midpoint.

Negative sharpness scores are outside this measurement domain. Equal-width extensions outside the valid domain are not collection targets.

Good includes both endpoints; the adjacent outer cutoffs belong to Needs improvement. Further tails are Bad in this scenario.

1 recordings lie outside the displayed five-band window; arrows show their IDs and values. All 30 enter the counts.

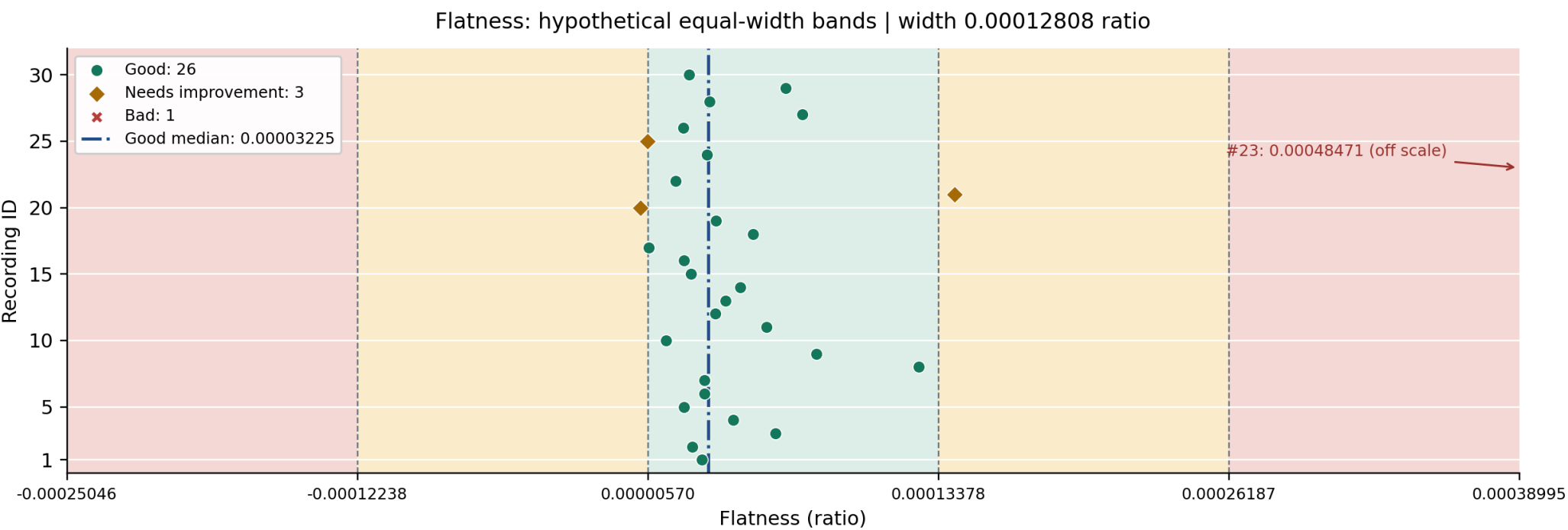
All source recordings were labeled good. These hypothetical classes need independent listener validation; production thresholds are unchanged.

Flatness: equal-width theoretical ranges

KaraOK / Hypothetical sampling scenario / 30 existing recordings / Units: ratio

Leftmost - Bad	Left - Needs improvement	Good	Right - Needs improvement	Rightmost - Bad
-0.00025046 to < -0.00012238	-0.00012238 to < 0.00000570	0.00000570 to 0.00013378	> 0.00013378 to 0.00026187	> 0.00026187 to 0.00038995

Five equal-width theoretical bands. The graph and counts use full-precision thresholds; displayed values are rounded.



Under these hypothetical boundaries: 26 Good / 3 Needs improvement / 1 Bad

Good-data median: 0.00003225 ratio, from the 26 recordings inside the Good range; not the range midpoint.

Valid measurement domain: 0 to 1 ratio. Equal-width extensions outside the valid domain are not collection targets.

Good includes both endpoints; the adjacent outer cutoffs belong to Needs improvement. Further tails are Bad in this scenario.

1 recordings lie outside the displayed five-band window; arrows show their IDs and values. All 30 enter the counts.

All source recordings were labeled good. These hypothetical classes need independent listener validation; production thresholds are unchanged.